D1 Differential calculus II: problems

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Unless stated otherwise, angles are in radians and "exact" means no decimals: keep π, e, $\ln 2$, $\sqrt{3}$ and so on. The point values in the CE-level section, such as (4p), show how many points a similar question would be worth on the central exam (CE = *centraal examen*). Use them to judge how much work an answer needs.

Practice

1. Differentiate (chain rule).

(a) $f(x) = (3x^2 - 5)^4$

(b) $g(x) = \sqrt{1 + x^2}$

(c) $h(x) = \dfrac{1}{(2x - 1)^3}$

(d) $k(x) = \sin(3x^2)$

2. Differentiate. Simplify where possible.

(a) $f(x) = x^2 e^{-x}$

(b) $g(x) = \ln(x^2 + 1)$

(c) $h(x) = x \cdot 2^x$

(d) $k(x) = {}^3\log(5x)$

(e) $m(x) = \tan(2x)$

(f) $n(x) = e^{\sin x}$

3. Given $f(x) = \dfrac{\ln x}{x}$ for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - \ln x}{x^2}$.

(b) Calculate exactly the coordinates of the extreme of f, and show it is a maximum.

(c) Use (b) to decide, without a calculator, which is larger: e^{π} or π^e. (Hint: compare $f(e)$ and $f(\pi)$.)

4. Let $f(x) = \sin^2 x + \cos^2 x$. Without using the identity $\sin^2 x + \cos^2 x = 1$, compute $f'(x)$ with the chain rule. What does your answer prove about f? What is the value of the constant?

5. (a) Prove that $\left({}^g\log x\right)' = \dfrac{1}{x \ln g}$ for $g > 0$, $g \neq 1$.

(b) Given $f(x) = {}^2\log(x^2 - 4)$. Calculate $f'(3)$ exactly.

6. The circle $x^2 + y^2 = 25$ passes through $P(3, -4)$.

(a) Use implicit differentiation to find the slope of the tangent line at P.

(b) Show that this tangent line is perpendicular to the radius OP.

7. The *folium of Descartes* is the curve $x^3 + y^3 = 6xy$.

(a) Show that $(3, 3)$ lies on the curve and find the equation of the tangent line there.

(b) Show that $\dfrac{dy}{dx} = \dfrac{x^2 - 2y}{2x - y^2}$.

(c) Calculate exactly the point with $x > 0$ where the tangent line is horizontal.

8. A spherical balloon is inflated at a rate of 100 cm³/s. How fast is its radius increasing at the moment the radius is 10 cm? Give an exact answer. ($V = \frac{4}{3}\pi r^3$.)

9. A ladder of 5 m leans against a vertical wall. The foot slides away from the wall at 0.5 m/s. How fast does the top slide down the wall when the foot is 3 m from the wall?

10. (a) Find the inflection points of $f(x) = x^4 - 6x^2 + 1$.

(b) Find the inflection point of $g(x) = xe^{-x}$.

11. The function $f(x) = x^3 + x + 1$ has an inverse function f^{-1}.

(a) Explain why f has an inverse.

(b) Calculate $(f^{-1})'(3)$ exactly.

CE-level (*centraal-examenniveau*)

12. Given $f(x) = x^2 \ln x$ for $x > 0$.

(a) Calculate exactly the coordinates of the extreme of f. (4p)

(b) Calculate exactly the coordinates of the inflection point (*buigpunt*) of f. (4p)

(c) *Toon aan* (show) that the tangent line at the inflection point intersects the y-axis at $\left(0, \dfrac{1}{2e^3}\right)$. (4p)

13. Given $f(x) = \sin x + \frac{1}{2}\sin(2x)$ on $[0, 2\pi]$.

(a) Show that $f'(x) = (2\cos x - 1)(\cos x + 1)$. (3p)

(b) Calculate exactly the maximum value of f. (4p)

(c) Show that $f'(\pi) = 0$ but that f has no extreme at $x = \pi$. (3p)

14. (a) *Toon aan dat de lijn* $y = ex$ *de grafiek van* $f(x) = e^x$ *raakt.* (3p)

(b) The line k passes through the origin and touches the graph of $g(x) = \ln x$. Calculate exactly the coordinates of the point of contact and give an equation of k. (4p)

15. For every p the function $f_p(x) = x^4 + px^2$ is given.

(a) Show that for $p \geq 0$ the function f_p has exactly one extreme. (3p)

(b) For $p < 0$, f_p has three extremes. Calculate the coordinates of the two minima in terms of p. (4p)

(c) Show that for $p < 0$ the two minima lie on the curve $y = -x^4$. (3p)

16. For every $p \neq 0$ the function $f_p(x) = xe^{-px}$ is given.

(a) Calculate the coordinates of the extreme of f_p in terms of p. (4p)

(b) Show that all these extremes lie on one straight line through the origin, and give its equation. (3p)

(c) Calculate p for which the inflection point of f_p lies at $x = 4$. (4p)

17. A cylindrical drinks can must hold 330 cm³. The material cost is proportional to the total surface area (side, top and bottom).

(a) Show that the surface area is $S(r) = 2\pi r^2 + \dfrac{660}{r}$, where r is the radius in cm. (3p)

(b) Calculate the radius that makes S minimal, in two decimals, and show that for this can the height equals the diameter. (5p)

18. Point P moves along the parabola $y = x^2$. Point A is $(0, 3)$. Calculate exactly the smallest possible distance AP and the points P where it is reached. (5p)

19. A water tank has the shape of an upside-down cone (point at the bottom) with height 4 m and top radius 2 m. Water flows in at 0.5 m³/min.

(a) Show that when the water is h m deep, the volume of water is $V = \dfrac{\pi h^3}{12}$. (3p)

(b) Calculate exactly how fast the water level rises at the moment the water is 1 m deep. (4p)

20. *Functieonderzoek.* Given $f(x) = \dfrac{x^2 + 3}{x - 1}$.

(a) Give the equations of all asymptotes of the graph of f. (4p)

(b) Calculate exactly the coordinates of the extremes, and say which is a maximum and which a minimum. (5p)

(c) Show that the graph has no inflection points. (3p)

(d) Give the range (*bereik*) of f. (2p)

21. Curve K is given by $x^2 - xy + y^2 = 3$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{y - 2x}{2y - x}$. (3p)

(b) Show that the tangent line to K at the point $(2, 1)$ is vertical. (3p)

(c) Calculate exactly the coordinates of the highest point of K. (4p)

22. Given $f(x) = x + 2\cos x$ on $[0, 2\pi]$.

(a) Calculate exactly the coordinates of the extremes (not the boundary extremes). (4p)

(b) Show that the tangent line at the inflection point with $x = \frac{\pi}{2}$ passes through $(\pi, 0)$. (4p)

Challenge

23. Let p and q be whole numbers with $q > 0$, and $y = x^{p/q}$ for $x > 0$. Use $y^q = x^p$ and implicit differentiation to prove that $y' = \dfrac{p}{q} x^{p/q - 1}$.

24. (a) Prove that $(\arctan x)' = \dfrac{1}{1+x^2}$.

(b) Let $h(x) = \arctan x + \arctan \dfrac{1}{x}$ for $x > 0$. Show that $h'(x) = 0$, and find the constant value of h.

(c) What is $h(x)$ for $x < 0$?

25. Let $f(x) = x^x$ for $x > 0$.

(a) Find $f'(x)$.

(b) Find the exact minimum value of f.

(c) What happens to $f(x)$ as $x \to 0^+$? (Hint: what happens to $x \ln x$?)

26. For which base $a > 1$ does the graph of $y = a^x$ touch the line $y = x$? Find the point of contact. Then explain for which $a > 1$ the equation $a^x = x$ has zero, one or two solutions.

27. The *astroid* is the curve $x^{2/3} + y^{2/3} = 1$. Take a point (x_0, y_0) on it with $x_0, y_0 > 0$.

(a) Show that the slope of the tangent line there is $-\left(\dfrac{y_0}{x_0}\right)^{1/3}$.

(b) Show that the tangent line meets the x-axis at $(x_0^{1/3}, 0)$ and the y-axis at $(0, y_0^{1/3})$.

(c) Conclude that the piece of every such tangent line between the axes has length 1. (A ladder of length 1 sliding down a wall sweeps out exactly this curve.)

28. Investigate, for every real p, how many solutions the equation $e^x = px$ has.

29. Let $f(x) = x^2 \sin \dfrac{1}{x}$ for $x \neq 0$ and $f(0) = 0$.

(a) Using the definition of the derivative, show that $f'(0) = 0$.

(b) Compute $f'(x)$ for $x \neq 0$.

(c) Show that $f'(x)$ does not tend to a limit as $x \to 0$. So f is differentiable everywhere, but f' is not continuous.

(d) Explain why this f is exactly the kind of inner function that breaks the simple proof of the chain rule in notes section 3.